

Summary

- AI/LLMs are becoming practical tools for biomedical research, coding, and bio-big data analysis, but their usefulness depends on the right context.
- Genomic data are massive: one human genome has about 3 billion base pairs, and NGS enabled massively parallel short-read sequencing.
- Whole-genome sequencing has become dramatically faster and cheaper, making large-scale genomic analysis and precision medicine feasible.
- Precision medicine moves beyond a one-size-fits-all approach toward the right drug, for the right patient, at the right dose.
- Global initiatives combine genomic, clinical, and cohort data to improve prevention, diagnosis, treatment, and pharmacogenomics.
- A central goal of human genetics is genotype-based prediction of phenotype through sequencing, annotation, and functional interpretation.
- AI can accelerate the management and interpretation of genomic big data and help translate bioinformatics into real-world precision medicine.